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Inventor(s)

Torres; Chuka

TRIMMER WITH MULTIPLE SETTINGS

Abstract

A head of the trimmer includes a fixed blade and vibrating blade joined together through a block in a manner allowing vibration of the vibrating blade relative to the fixed blade. A block is attached to the vibrating blade in a removable fashion, and resides within a slot which allows the vibrating blade and block to slide relative to the fixed blade and remain connected to the fixed blade. The block can be removed, flipped and reattached, with the block being provided with top and bottom offsets which are different in size from each other, such that flipping of the block causes a tip line of teeth of the vibrating blade to have a different offset with a tip line of teeth of the fixed blade. The trimmer can thus be reconfigured between a standard configuration and a tight configuration, the tight configuration cutting hair shorter than the standard configuration.

Inventors: Torres; Chuka (West Hollywood, CA)

Applicant: Torres; Chuka (West Hollywood, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit under Title 35, United States Code § 119(e) of U.S. Provisional Application No. 63/559,508 filed on Feb. 29, 2024.

FIELD OF THE INVENTION

[0002] The following invention relates to electrically powered trimmers and clippers for cutting hair. More particularly, this invention relates to haircutting trimmers and clippers which can be adjusted between at least two settings, which settings cut the hair to different lengths.

BACKGROUND OF THE INVENTION

[0003] Hair trimmers are known in the prior art for cutting hair. Typical hair trimmers are electric powered and can be either corded or cordless. Hair trimmers include a head assembly with a fixed blade and a vibrating blade. The first fixed blade has a series of teeth at a tip thereof. The second vibrating blade is adjacent to the fixed blade and also has a series of teeth at a tip thereof. The vibrating blade moves laterally relative to the fixed blade, causing the teeth of the fixed blade and the vibrating blade to move into and out of alignment with each other. Lateral sides of these teeth are sharpened so that they act somewhat like scissors and cut hair passing between these teeth.

[0004] Typically, the teeth of the vibrating blade are aligned along a straight line at their tip which extends slightly less away from a main body of the vibrating blade than a line along which tips of the teeth on the fixed blade are located. This small offset allows the teeth of the fixed blade to act somewhat like a “standoff” to keep the teeth from cutting skin from which the hair to be cut is growing.

[0005] It is known in the prior art to “set” the position of the vibrating blade relative to the fixed blade to control this amount of offset. A typical regular setting for a trimmer is to have just enough offset so that the risk of cutting skin is essentially zero. In this regular setting, a barber (or the individual cutting hair) does not need to exercise extra care to avoid cutting skin. However, as a trade-off, hair cannot be cut quite as short in such a regular setting.

[0006] Skilled barbers will often set their vibrating blade position to a somewhat tighter position than the regular setting. With such a tighter setting, the hair can be cut shorter and with skill and experience a trimmer with such a tight blade setting can still be safely used.

[0007] With prior art trimmers the process of setting the height of the vibrating blade typically involves loosening screws or other fasteners which hold the vibrating blade. Then the vibrating blade is positioned where desired, and finally the screws or other fasteners are tightened to cause the trimmer to hold this new setting. To get the setting right, tools can be utilized. One such tool is known as a blade setter and is similar to that shown and described in U.S. Pat. No. D785,249, incorporated herein by reference and by including a copy within the Appendix of this application.

[0008] Even with tools to assist in setting the blade where desired, significant time (and care) is involved in performing this setting procedure. Furthermore, if the setting procedure is not done properly the trimmer can be left in a state which is suboptimal. While a barber could maintain two trimmers, one at a regular (or standard) setting and one at a tight setting, this is wasteful of trimmer resources. Accordingly, a need exists for a trimmer which can be rapidly and reliably transitioned between a regular setting and a tight setting.

SUMMARY OF THE INVENTION

[0009] With this invention, a hair trimmer is provided which has at least two settings, such as a regular (or standard) setting and a tight setting. In one embodiment disclosed in the included figures, a base of the fixed blade is provided with a dovetail slot which extends laterally generally parallel with a line at which tips of the teeth of the fixed blade are oriented. This dovetail slot in this embodiment extends entirely from a first lateral edge to a second lateral edge of the fixed

blade, but could stop short of these lateral edges as an option. In other embodiments, this slot could have a different contour.

[0010] A proximal end of the base of the fixed blade (and of the overall head assembly) preferably includes a pair of retainer seats therein. These retainer seats are sized and positioned to allow spring ball retainers to reside therein and to removably hold the head assembly attached to a neck of a trimmer handle. The handle is configured within a hand grip sized case in one embodiment, similar to the handle shown in U.S. Patent Application Publication No. 2023/0278243, incorporated herein by reference in its entirety. The handle can be easily gripped and wielded by a barber or other user. This handle can house a battery or can support an electric cord attachable to a source of electric power. A vibrating motor within the handle has an output which interacts with horns (or a collar) of the head assembly and which causes the vibratory blade of the head assembly to vibrate relative to the fixed blade when the head assembly is attached to the neck of the handle and the vibratory output tip of the output of the vibrating motor engages these horns. The neck also supports a pair of spring ball retainers which are spring loaded towards an elevated position, but can be depressed when compressive forces are applied to these balls. The balls can seat within the spring ball retainer seats in the proximal end of the base of the fixed blade, which proximal end also provides a proximal end for the entire head assembly.

[0011] The vibrating blade is provided as part of the head assembly and is configured to vibrate back and-forth under action from the tip of the output of the vibrating motor. The fixed blade and the proximal end of the head assembly are held stationary by the ball retainers and the neck of the handle. The proximal end of the head assembly includes flanges which engage within slots in the neck of the handle to allow for the head assembly to slide into and be firmly held by the neck of the handle, and resist sliding and falling out of the neck by the spring force of the spring ball retainers nesting at least somewhat into the retainer seats in the proximal end of the head assembly.

[0012] A block is provided between the vibrating blade and the fixed blade. This block, in one embodiment, resides within a corresponding slot of the fixed blade. The block has at least one raised region sized to fit within a recess in the vibrating blade. Block fasteners allow for the block to be fastened to the vibrating blade, such as through a base block on a side of the vibrating blade opposite of the fixed blade. The block fasteners (typically two screws, but could optionally be rivets or other permanent fasteners) securely but removably hold the vibrating blade to the block. In alternative embodiments, the block could have a contour other than a dovetail contour.

[0013] The block has a size at the raised region (in a direction perpendicular to a tip line at which tips of the teeth terminate) which is less than a size of the recess in the vibrating blade, by at least a defined adjustment amount. This causes a small gap to exist adjacent to at least one side of the raised region. In one embodiment, this adjustment amount is about 0.1 mm. Thus for instance, the size of the raised region of the block would be at least about 0.1 mm less than the size of the recess. In other embodiments, this tolerance and adjustment amount could be in a range from 0.1 mm to 0.5 mm, either built into the trimmer or selected by a user, such as by using a desired block from a set of different size blocks.

[0014] The block can be removed from the vibrating blade, after the vibrating blade and the block are slid off of the fixed blade. Such removal first involves removing fasteners which hold the block to the vibrating blade. The block can then be flipped and rotated with its at least one raised region fitting again within the recess. By making the raised regions asymmetrical and/or by putting a through bore in the block (that has the screw or other fastener passing therethrough) and a threaded bore of the vibrating blade asymmetrically located relative to the dovetail of the block, the block causes the vibrating blade to be in a standard or a tight position, depending on if the block is in a standard orientation or a tight orientation.

OBJECTS OF THE INVENTION

[0015] Accordingly, a primary object of the present invention is to provide a trimmer which has both a standard configuration and a tight configuration with the tight configuration cutting hair

shorter than the standard configuration.

[0016] Another object of the present invention is to provide a trimmer which can be reconfigured either to accommodate sensitive skin or a less experienced user in a first configuration and provide a shorter cut and tighter hairline shaping by a skilled user when in a second configuration.

[0017] Another object of the present invention is to provide a trimmer with a standard and a tight configuration which is quick and easy to reconfigure when desired.

[0018] Another object of the present invention is to provide a trimmer with a tight configuration and a standard configuration which operates similarly in each configuration, but merely with a slightly different offset between tip lines of fixed and vibrating teeth in the two configurations.

[0019] Another object of the present invention is to balance safety and comfort with exceptionally high quality hairline shaping in a single trimmer tool.

[0020] Another object of the present invention is provide a hair trimmer tool which maintains its offset between fixed teeth and vibrating teeth, even after long periods of use thereof, with a minimum of wear or other performance degradation.

[0021] Another object of the present invention is to provide a method for reconfiguring a hair trimmer between a standard configuration and a tight configuration so that a single tool can provide two different configurations to a user and a simple and fast method for reconfiguring the hair trimmer tool between configurations.

[0022] Other further objects of the present invention will become apparent from a careful reading of the included drawing figures, the claims and detailed description of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is an exploded parts perspective view of a trimmer tool according to one embodiment of this invention, and showing a handle of the tool exploded into separate parts and a head of the tool exploded into separate parts.

[0024] FIG. 2 is a perspective view similar to that which is shown in FIG. 1, but with the handle assembled together and the head assembled together, but with the head exploded off of the handle as it would be when removed from a neck of the handle.

[0025] FIG. 3 is a perspective view of the tool when the head is attached to the neck of the handle and ready for use.

[0026] FIG. 4 is a bottom plan view of that which is shown in FIG. 3.

[0027] FIG. 5 is a left side elevation view of that which is shown in FIG. 3, the right side elevation view being a mirror image of that which is shown in FIG. 5.

[0028] FIG. 6 is a top plan view of that which is shown in FIG. 3.

[0029] FIG. 7 is an upper end elevation view of that which is shown in FIG. 3.

[0030] FIG. 8 is a base end elevation view of that which is shown in FIG. 3.

[0031] FIG. 9 is a perspective view from below of the head portion of that which is shown in FIGS. 1-3.

[0032] FIG. 10 is a top plan view of that which is shown in FIG. 9.

[0033] FIG. 11 is a left side elevation view of that which is shown in FIG. 9.

[0034] FIG. 12 is a bottom plan view of that which is shown in FIG. 9.

[0035] FIG. 13 is a right side elevation view of that which is shown in FIG. 9.

[0036] FIG. 14 is a perspective view of the head of FIG. 9 with the vibrating blade slid off of the fixed blade and with a dovetail block and screws exploded away from the vibrating block as they would be when removing the screws and dovetail block for a first step in reconfiguring of the head between a standard configuration and a tight configuration.

[0037] FIG. 15 is a perspective view similar to that which is shown in FIG. 14, but with the

dovetail block beginning a flipping procedure to reorient the vibrating blade relative to the fixed blade for reconfiguring the head between the standard configuration and the tight configuration. [0038] FIG. **16** is a perspective view similar to that which is shown in FIG. **15**, but when the dovetail block has nearly completed its flipping procedure.

[0039] FIG. **17** is a perspective view similar to that which is shown in FIG. **16**, but with the dovetail block having been fully flipped and mostly repositioned relative to the vibrating block and ready for receiving the screws to reattach the dovetail block to the vibrating block before sliding the vibrating blade and dovetail block back into the dovetail slot of the fixed blade, for completion of the process of reconfiguring the head between the tight configuration and the standard configuration. FIG. **17** also shows an alternative embodiment for the vibrating blade with a set screw as one form of tension adjustment between the vibrating blade and the fixed blade.

[0040] FIG. **18** is a perspective view of a detail of the head after full reassembly and with the teeth of the head in a tight configuration.

[0041] FIG. **19** is a perspective view of that which is shown in FIG. **18**, but with the teeth of the head in a standard configuration before the reconfiguration process illustrated in FIGS. **14-17**.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0042] Referring to the drawings, wherein like reference numerals represent like parts throughout the various drawing figures, reference numeral **10** is directed to a trimmer for cutting hair (FIGS. **1-3**). The trimmer **10** can be reconfigured between a tight configuration (FIG. **18**) and a standard configuration (FIG. **19**), with the tight configuration cutting hair shorter than the standard configuration. A user can thus configure the trimmer **10** in a desired way for either a standard cut or a tight cut.

[0043] In essence, and with particular reference to FIG. **1**, basic details of trimmers of this invention are described, according to one example embodiment. The trimmer **10** includes a handle **20** (FIGS. **2** and **3**) having a hand grip size and shape. A motor **30** is contained within the handle **20** and drives vibratory operation of a head **40**, which is removably attachable to the handle **20** (along arrow A of FIG. **2**).

[0044] The head includes a fixed blade **50** and vibrating blade **60** which are held together in a fixed relationship other than facilitating vibration of the vibrating blade **60** back-and-forth relative to the fixed blade **50** along a vibrating movement (arrow E of FIGS. **3**, **18** and **19**) parallel with tips **51**, **61** of the blades **50**, **60**. A block **70** joins the vibrating blade **60** to the fixed blade **50**. The block **70** can be removably attached, such as through screws **80**, to the vibrating blade **60** in two different orientations which correspond with the standard configuration (FIG. **19**) and the tight configuration (FIG. **18**) of the trimmer **10**.

[0045] More specifically, and with continuing reference to FIGS. **1-3**, details of the handle **20** and motor **30** of the trimmer **10** are described, according to one example embodiment. In this embodiment, the handle **20** is formed of two separate housing halves **22** which fit together in a clamshell manner to enclose all of the handle **20** other than an opening at a neck **25** at an upper end of the handle **20**. Handle **20** supports charging contacts should the trimmer **10** be of a rechargeable cordless style, or can support a strain relief and cord if of a corded style.

[0046] A motor **30** is supported within the handle **20** for powering the trimmer **10**. This motor **30** is typically coupled to a battery which is also contained within the handle **20**. A switch **24** is provided on the handle **20**, which allows for opening and closing of a circuit with battery contacts, for turning the motor **30** on and off. The neck **25** includes flanges **27** on lateral sides thereof slightly above a shelf **26** which is parallel with but recessed below the flanges **27** slightly. The flanges **27** extend inwardly toward each other somewhat on opposing sides of the neck **25**. The shelf **26** supports a pair of spring ball retainers **28** which extend up toward a plane in which the neck **25** is located just enough that the spring balls interfere with the head **40** of the trimmer **20** when it is slid onto and off of the flanges **27** of the neck **25** (along arrow A of FIG. **2**).

[0047] The head **40** includes a pair of recesses **55** in the fixed blade **50** of the head **40** which are

spaced apart a similar distance that the spring ball retainers **28** are spaced from each other. The recesses **55** are strategically located so that when the head **40** is slid onto the flanges **27** on lateral sides of the neck **25** of the handle **20**, along arrow A of FIG. **2**, the recesses **55** come into alignment with the spring ball retainers **28**. The spring ball retainers **28** are biased toward a position with the balls elevated to block passage of the head **40** between the flanges **27** and onto the neck **25**. However, these balls of the spring ball retainers **28** can be collapsed slightly to allow the head **40** to pass fully onto the neck **25** of the handle **20**, and then the balls are able to snap into the recesses **55** and hold the head **40** onto the handle **20**.

[0048] A firm force opposite arrow A applied to the head **40** relative to the handle **20** allows for the balls to be compressed into the spring ball retainers **28** and out of the recesses **55**, sufficient to allow the head **40** to be slid off of the handle **20** when desired (such as when it is desired to reconfigure the head **40** to transition the trimmer **10** from a standard configuration (FIG. **19**) into a tight configuration (FIG. **18**). A trapezoidal post **29** is also preferably provided extending up from the shelf **26** which can align with portions of the head **40** to further aid in securing the head **40** in a removable fashion onto the handle **20**, and so that the head **40** remains firmly fixed to the handle **20** during use of the trimmer **10**.

[0049] The motor **30** is an electric motor which receives electric power from a battery or from some other electric power source, such as that supplied through a cord interfacing with the motor **30** through the handle **20**, and also with the switch **24** on the handle **20**. The motor **30** has an output shaft on an upper end thereof which is fitted with an eccentric output tip **32**. This eccentric output tip **32** has a rotational axis aligned with an output shaft of the motor **30**, but a portion of the eccentric output tip **32** most elevated above the motor **30** is offset from this rotational axis slightly (e.g. 5 mm). Thus, this elevated portion follows a tight circular path as it rotates (along arrow B of FIG. **1**) when the motor **30** is operational.

[0050] This elevated portion of the eccentric output tip **32** resides within a collar **64** associated with the vibrating blade **60**, which collar is caused to slide back-and-forth (along arrow E of FIGS. **3**, **18** and **19**) when the head **40** is attached to the handle **20** and the motor **30** is operating. A counter balance **34** is also preferably attached to an output shaft of the motor **30** which counter balance **34** has a center of mass opposite a center of mass of the eccentric output tip **32** of the motor **30**. In this way, the output shaft of the motor **30** is balanced, but still has an elevated portion which revolves about the rotational axis of the motor, to cause vibrational motion (along arrow E) of the vibrating blade **60**. Typically, various electronics or other electric circuitry are also coupled to the motor **30** as well as to a battery and/or electric cord and the switch **24**.

[0051] With particular reference to FIGS. **1**, **2** and **9-14**, details of the head **40** are described, according to this example embodiment. The head **40** provides a cutting implement for the trimmer **10** through interaction of teeth **53**, **63** associated with the fixed blade **50** and vibrating blade **60** moving relative to each other (along arrow E of FIGS. **3**, **18** and **19**). The head **40** includes a gap **42** provided within a base block **52** attached to the fixed blade **50**. This gap **42** is provided on each lateral side of the fixed blade **50** of the head **40** and is sized to allow the flanges **27** of the neck **25** to slide therein. In this way, the gap **42** provides for sliding attachment and detachment of the head **40** onto and off of the neck **25** of the handle **20**. An outer rim **58** of the fixed blade **50** preferably overlays the neck **25** slightly and defines a perimeter of the head **40**, which is visible when the head **40** is attached to the neck **25** of the handle **20**.

[0052] The fixed blade **50** is thus attached through its base block **52** securely to the neck **25** of the handle **20**, when the head **40** is attached to the handle **20**, and without any motion between the fixed blade **50** and the handle **20**. In contrast, the vibrating blade **60** moves relative to the fixed blade **50** and relative to the handle **20**. The fixed blade is a rigid structure, typically formed of a tool steel which can hold a sharp edge, such as along lateral sides of teeth **53** extending to a tip **51** of the fixed blade **50**. The tip **51** is defined by a tip line to which each of the teeth **53** of the fixed blade **50** extend. This tip line **51** defines a portion of the head **40** which extends furthest away from

the base block **52** of the fixed blade **50**.

[0053] While this tip **51** always extends to this tip line, the vibrating blade **60** also has a tip line defined by tips **61** of teeth **63** on the vibrating blade **60**, which are close to the tip line defined by the tips **51** of the teeth **53** on the fixed blade **50**. An offset between the tip **51** of the blade **50** and the tip **61** of the vibrating blade **60** defines whether the trimmer **10** is in its standard configuration (FIG. **19**) or its tight configuration (FIG. **18**). In one embodiment, the standard configuration involves an offset of 0.2 mm, and the tight configuration involves an offset of 0.1 mm. However, in other embodiments other measurements for any such two configurations could be provided by keeping the tips **51** of the fixed blade **50** extending furthest within the head **40**. Potential for the teeth **53**, **63** to interact in a way cutting skin of a user is thus substantially precluded, even in the tight configuration. However, the standard configuration provides a greater margin of safety and is often more appropriate for less skilled users and for sensitive skin. When it is desirable to cut the hair as short as possible, the tight configuration is most effective.

[0054] Various tools can be utilized to sharpen the teeth **53**, **63** and to otherwise modify the teeth **53**, **63** to adjust a tightness provided by the trimmer **10** in the two configurations. One such sharpening and modifying tool is described in U.S. Pat. No. 10,556,324, incorporated herein by a reference in its entirety. With some trimmers **10**, a spacing between tips **51**, **61** of the fixed blade **50** and vibrating blade **60** can to some extent be modified, such as by utilizing a modifier tool, such as that disclosed in U.S. Pat. No. D785,249, incorporated herein by reference in its entirety.

[0055] The fixed blade **50** also includes a trapezoidal slot **57** within the base block **52** which can correspond with and be attached to a trapezoid post **29** in the shelf **26** of the handle **20**, which further allows for tight fitting and attachment of the fixed blade **50** of the head **40** to the handle **20** without vibration or other movement between the head **40** and handle **20**. By making the trapezoidal slot **57** and trapezoid post **29** with tapering side walls, they maintain a tight hold to each other when the head **40** is attached to the handle **20**.

[0056] The fixed blade **50** includes an undersurface **54** upon which the base block **52** is attached, typically through screws. As an option, the base block **52** could be formed with the fixed blade, **50** or attached in some manner other than threaded fasteners, such as an adhesive, rivets or other mechanical fasteners. The undersurface **54** features a dovetail slot **56**, which extends along an entire lateral width of the fixed blade **50** in this embodiment. A center line of the dovetail slot **56** is parallel with the tip line defined by the tips **51** of the teeth **50** of the fixed blade **50**.

[0057] This dovetail slot **56** has tapering side walls, giving it a dovetail appearance, in this example embodiment. In other embodiments, at least one of the two sides of the slot **56** taper, but not necessarily both of them. In another embodiment, rather than angled or beveled side surfaces for the dovetail slot **56**, these side edges could be stepped and still allow for structures associated with the vibrating blade **60** to slide laterally, without being able to be displaced in any manner relative to the dovetail slot **56** other than in this lateral direction (along arrows E and F (FIGS. **3**, and **17-19**)).

[0058] Vibrating blade **60** is slightly smaller than the fixed blade **50** and similarly formed of a rigid material, typically of metal (e.g. tool steel), which can have its teeth **63** sharpened and maintain a sharp character, able to act with the teeth **53** of the fixed blade **50** for cutting hair therebetween. The teeth **63** extend to tips **61** which define a tip line which is parallel with the tip line associated with the tips **50** of the fixed blade **50**. However, the tips **61** of the vibrating blade **60** are held back slightly to extend not quite as far as the tips **51** of the fixed blade **50**.

[0059] A base block **62** attaches to the vibrating blade **60** on a side opposite the fixed blade **50**. This base block **62** could be formed with the vibrating blade **60**, but typically is attached to the vibrating blade **60** through threaded fasteners or pressfit studs or rivets. In one embodiment, these pressfit studs each have a threaded bore **65** within an interior thereof which receives the screws **80** which hold the dovetail block **70** to the vibrating blade **60**.

[0060] The base block **62** of the vibrating blade **60** supports a collar **64** at a center thereof which is located adjacent to the eccentric output tip **32** of the motor **30** when the head **40** is attached to the

neck **25** of the handle **20**. The eccentric output tip **32** resides inside of the collar **64** (or other guide rails) and impacts the collar **64** as the eccentric output tip rotates (along arrow B of FIG. 1). Because the vibrating blade **60** is restrained from any motion relative to the fixed blade **50** and the handle **20** other than lateral back-and-forth motion (along arrow E of FIGS. 3, 18 and 19), the interaction between the eccentric output tip **32** and the collar **64** results in only vibrating back-and-forth motion of the vibrating blade **60** relative to this fixed blade **50**.

[0061] An inside surface **66** of the vibrating blade **60** between the base block **62** of the vibrating blade **60** and the tips **61** of the teeth **63** is provided with a rectangular recess **68**. While this recess **68** has a rectangular cross-section in the embodiment shown, other cross-sections could conceivably be alternatively provided, including a square cross-section, or a trapezoid cross-section, or a stepped cross-section. With such a rectangular cross-section, at least one raised region **76** of the dovetail block **70** can most readily be inserted into this rectangular recess **68** for attachment of the dovetail block **70** in removable fashion to the vibrating blade **60**.

The recess **68** is oversized relative to the at least one raised region **76**, to allow the block **70** to be asymmetrical and still fit in the recess **68** in either of two orientations (FIGS. 18 and 19). The through bore **78** is slightly off center relative to the dovetails **74**, so that this asymmetry is provided, and the recess **68** accommodates both orientations, with a small gap either at the top offset **77** or at the bottom offset **77** or at the bottom offset **79**, but so slight (e.g. 0.1 mm) that it does not show in the drawings. The rectangular recess **68** preferably extends along an entire lateral width of the vibrating blade **60**, but could conceivably stop short of extending entirely between lateral edges of the vibrating blade **60**.

[0062] The dovetail block **70** is provided which is removably attachable to the vibrating blade **60**, and which resides partially within the dovetail slot **56** of the fixed blade **50** for attachment to the fixed blade **50**, but in a manner which allows sliding of the dovetail block **70** and attached vibrating blade **60** relative to the fixed blade **50** (along arrow E of FIGS. 3, 18 and 19). This dovetail block **70** is preferably a rigid mass of material which includes a first side **71** opposite of second side **72**. One of these sides has the at least one raised region **76** (and typically two such raised regions **76**), which are sized and shaped to fit within the at least one recess **68** in the inside surface **66** of the vibrating blade **60**.

[0063] The threaded bores **65** in the vibrating blade **60** and base block **62** are aligned with through bores **78** in the dovetail block **70**, which also pass through these rectangular raised regions **76**, to receive screws **80** therethrough and for attachment of the dovetail block **70** to the vibrating blade **60** within the rectangular recesses **68** thereof. While the dovetail block **70** is referred to this way because it includes a dovetail **74** on a side thereof opposite the rectangular raised regions **76**, the dovetail **74** could be replaced with some other structure which would have a complementary fit within the dovetail slot **56** of the fixed blade **50**. For instance, if the dovetail slot **56** were instead configured with a stepped form, the dovetail **74** of the dovetail block **70** could also be provided with a corresponding step form. At a minimum, preferably the slot **56** has a cross-section (perpendicular to the tip line of the teeth **53**) with a similar size at an outer part of the slot than at a floor of the slot. The dovetail **74** of the dovetail slot **70** can slide within the dovetail slot **76** in a lateral fashion parallel with the tip lines of the tips **51**, **61**, to facilitate vibrating of the vibrating blade **60** relative to the fixed blade **50**, but while keeping the vibrating blade **60** tight against the fixed blade **50**.

[0064] To facilitate ease in sliding the dovetails **74** into the dovetail slot **56**, dovetail tips **75** (FIG. 17) can be formed of a more flexible material than other parts of the dovetails **74** and/or the dovetail tips **75** can have their edges beveled. At least one of the dovetail tips **75** at each end of the dovetail block **70** would be configured as an option to facilitate a user sliding the dovetail block **70** into the dovetail slot **54**.

[0065] Importantly, the raised regions **76** of the dovetail block **70** have a top offset **77** and a bottom offset **79** on lateral sides thereof which are slightly different in size. The top offset **77** is slightly

lesser in size than the bottom offset **79**. In this way, when the top offset **77** is on a side of the rectangular slot **76** of the vibrating blade **60** closest to the tip **61** of the vibrating blade **60**, the vibrating blade **60** is caused to be in its tight configuration relative to the fixed blade **50**, when the dovetail block **70** is attached to the vibrating blade **60** in this manner and has its dovetail **74** fitted within the dovetail slot **56** of the fixed blade **50** (FIG. **18**). However, when the dovetail block **70** is removed from the vibrating blade **60**, reoriented in an opposite orientation and reattached to the rectangular slot **68** of the vibrating blade **60**, so that the bottom offset **79** is closer to the tips **61** of the teeth **60** through the vibrating blade **60**, the vibrating blade is caused to have its teeth **63** slightly further from the tips **51** of the teeth **53** of the fixed blade **60**, putting the head **40** of the trimmer **10** in its standard configuration (FIG. **19**).

[0066] It can be seen that different offsets **77**, **79** of different sizes could be provided so that differing distances are provided between the tight configuration and the standard configuration in alternative embodiments. As a further option, additional dovetail blocks could be provided in a kit along with the dovetail block **70**, which could have different offsets, so that different offsets between the tips **51** of the teeth **53** of the fixed blade and the tips **61** of the teeth **63** of the vibrating blade **60** could be provided when such alternative dovetail blocks are used to replace the dovetail block **70**. For instance, a dovetail block could be provided which provides an offset of 0.3 mm or 0.4 mm, depending on the orientation that it has when attached to the vibrating blade **60**. Preferably the vibrating blade **60** and/or dovetail block **70** are provided with indicia thereon which are visible and which indicate to a user whether the dovetail block **70** is oriented to provide a tight configuration or a standard configuration.

[0067] The screws **80** (FIG. **17**) can be tightened extra (or not) to enhance (or not) friction between the fixed blade **50** and the vibrating blade **60**. Furthermore, as an option, a set screw **90** (FIG. **17**) can be provided within a threaded hole **92** in a protrusion **94** extending from the base block **62** of the vibrating blade **60**. The set screw **90** can be tightened (or loosened) along arrow G to interact with the threaded hole **92** and be pushed into an outside surface **67** of the vibrating blade **60** (along arrow H of FIG. **17**). In this way, tension and/or friction between the blades **50**, **60** can be further tuned by a user. In one embodiment, this set screw **90** has a faceted hole that is accessible by a torque applying tool even when the head **40** is attached to the handle **20**, by this set screw **90** being located just outside of the neck **25** of the handle **20**.

[0068] In use and operation, and with particular reference to FIGS. **14-17**, a series of steps in the process of re-orienting the dovetail block **70** are shown, according to one example embodiment. Initially, the head **40** is assembled together with the vibrating blade **60** adjacent to the fixed blade **50** and with the dovetail block **70** joining the vibrating blade **60** to the fixed blade **50**. When the head **40** is removed from the handle **20** of the tool **10** (FIG. **2**) the vibrating blade **60** and attached dovetail block **70** can be slid laterally entirely off of the fixed blade **50** (along arrow F of FIG. **14**). Once the vibrating blade **60** and associated dovetail block **70** have been removed from the fixed blade **50**, the screws **80** can have heads thereof accessed, and they can be removed from the dovetail block **70** and vibrating blade **60** (along arrow C of FIG. **14**).

[0069] The dovetail block **70** can then be rotated/flipped (along arrow D of FIGS. **15** and **16**) to swap ends of the dovetail block **70** and cause it to be reoriented 180° from its original configuration to a final configuration (FIG. **17**). Dovetail block **70**, after such reorientation and flipping, can then have the screws **80** reattached (along arrow C of FIG. **17**) to reattach the dovetail block **70** to the vibrating blade **60**. The vibrating blade **60** and dovetail block **70** can then together be slid back into the dovetail slot **56** of the fixed blade **50** (along arrow F of FIG. **17**) to reattach the vibrating blade **60** to the fixed blade **50**, in a manner precluding motion therebetween, other than lateral vibrating motion (along arrow E of FIGS. **3**, **18** and **19**). Depending on which orientation the dovetail block **70** has within the rectangular recess **68** of the vibrating blade **60**, the tips **61** of the teeth **63** are either positioned in a tight configuration (FIG. **18**) with a 0.1 mm offset (for instance) or with a standard offset (FIG. **19**), with a 0.2 mm (for instance) offset between the teeth **63** of the vibrating

blade **60** and the teeth **53** of the fixed blade **50**. The head **40** can then be reattached to the handle **20** (along arrow A of FIG. 2) and the tool **10** is ready for use in the new configuration.

[0070] This disclosure is provided to reveal a preferred embodiment of the invention and a best mode for practicing the invention. Having thus described the invention in this way, it should be apparent that various different modifications can be made to the preferred embodiment without departing from the scope and spirit of this invention disclosure. When embodiments are referred to as “exemplary” or “preferred” this term is meant to indicate one example of the invention, and does not exclude other possible embodiments. When structures are identified as a means to perform a function, the identification is intended to include all structures which can perform the function specified. When structures of this invention are identified as being coupled together, such language should be interpreted broadly to include the structures being coupled directly together or coupled together through intervening structures. Such coupling could be permanent or temporary and either in a rigid fashion or in a fashion which allows pivoting, sliding or other relative motion while still providing some form of attachment, unless specifically restricted.

Claims

- 1:** A trimmer for cutting hair, the trimmer comprising in combination: a handle; a head assembly attached to the handle; said head assembly including a fixed blade and a vibrating blade; said fixed blade having a plurality of teeth extending to a common fixed blade tip line; said vibrating blade having a plurality of teeth extending to a common vibrating blade tip line; a vibrating output causing said vibrating blade to vibrate relative to said fixed blade in an at least partially lateral direction relative to said fixed blade tip line; said vibrating blade having at least two positions, including a regular position and a tight position, said tight position locating said vibrating blade tip line closer to said, fixed blade tip line than said regular position; and said vibrating blade movable between said regular position and said tight position.
- 2:** The trimmer of claim 1 wherein said handle includes a neck, said head assembly removably attachable to said neck of said handle.
- 3:** The trimmer of claim 2 wherein said vibrating output is supported by said handle and extends out of said neck for engagement with at least a portion of said head assembly.
- 4:** The trimmer of claim 1 wherein said fixed blade includes a slot on a side of said fixed blade facing said vibrating blade and extending parallel with said fixed blade tip line.
- 5:** The trimmer of claim 4 wherein said slot extends an entire width of said fixed blade.
- 6:** The trimmer of claim 4 wherein said slot has a cross-section with a smaller size at an outer part of said slot than at a floor of said slot.
- 7:** The trimmer of claim 6 wherein said vibrating blade includes a dovetail block residing partially within a recess in said vibrating blade through at least one raised region of said block sized to have a width perpendicular to said tip line of said vibrating blade which is less than a width of said recess perpendicular to said tip line, and wherein said block is also held partially within said slot of said fixed blade in a manner allowing vibrating motion parallel with said tip lines between said vibrating blade and said fixed blade.
- 8:** The trimmer of claim 7 wherein said block can have said raised region fit within said recess of said vibrating blade in two orientations rotated 180° from each other, including a standard orientation and a tight orientation, and with said raised region closer to said tip line of said vibrating blade when in said tight orientation than when in said standard orientation.
- 9:** The trimmer of claim 8 wherein said block includes a dovetail on a side of said block opposite said at least one raised region, and wherein said fixed blade includes said slot having dovetail corners therein, said slot sized to receive said dovetail of said block therein, said slot and said dovetail having constant cross-sectional forms to allow movement of said block within said slot of said fixed blade along a vibrating path parallel with said tip line of said fixed blade.

10: The trimmer of claim 9 wherein said block is attached to said vibrating blade through at least one removable fastener.

11: The trimmer of claim 10 wherein said block is attached to said vibrating blade through a pair of screws, passing along through bores in said block and extending into threaded bores in said vibrating blade, and wherein said through bores are located asymmetrically relative to said dovetail.

12: A hair trimmer, comprising in combination: a handle; a head removably attachable to said handle; said head including a fixed blade and a vibrating blade with said fixed blade fixed to said handle when said head is attached to said handle and said vibrating blade vibrating relative to said fixed blade; said fixed blade and said vibrating blade each having a plurality of teeth extending to blade tip lines for each of said blades, said blade tip lines each parallel with a direction of vibration of said vibrating blade relative to said fixed blade; said blade tip lines offset from each other with said blade tip line of said vibrating blade stopping short of said blade tip line of said fixed blade by an offset distance; a block between said fixed blade and said vibrating blade, said block holding said fixed blade to said vibrating blade in a manner allowing only lateral vibrating linear motion between said fixed blade and said vibrating blade; and said block movable between a standard orientation and a tight orientation, with said tight orientation causing said offset distance to be smaller than when said block is in said standard orientation.

13: The hair trimmer of claim 12 wherein said standard orientation and said tight orientation for said block are selectable by rotating said block between said standard orientation and said tight orientation.

14: The hair trimmer of claim 13 wherein said block is rotatable 180° to move between said standard orientation and said tight orientation.

15: The hair trimmer of claim 13 wherein said block includes a dovetail on a side thereof, said dovetail having a constant cross-sectional form, and wherein said fixed blade includes a dovetail slot therein sized similar to said dovetail of said block, said dovetail slot having a constant cross-sectional form extending parallel with said tip line of said fixed blade and with said block removably attachable to said vibrating blade.

16: The hair trimmer of claim 15 wherein said block includes at least one raised region on a side of said block opposite said dovetail, said raised region sized to fit within a recess in said vibrating blade, and wherein at least one fastener holds said block to said vibrating blade in a removable fashion.

17: The hair trimmer of claim 16 wherein said at least one raised region has a rectangular cross-section, and wherein said recess in said vibrating blade has a rectangular cross-section, with said at least one raised region fitting within said recess of said vibrating blade.

18: The hair trimmer of claim 16 wherein said block has at least one through bore for at least one fastener which through bore asymmetrically located relative to said dovetail, and with said block placing said vibrating blade within said tight orientation when said block is oriented with said through bore further from said tip line of said vibrating blade than when said block is oriented placing said vibrating blade within said standard orientation with said through bore closer to said tip line.

19: The hair trimmer of claim 16 wherein said fastener includes at least one screw passing along said at least one through bore passing through said block and threaded into a threaded bore in said vibrating blade.

20: A method for switching a hair trimmer between a standard orientation and a tight orientation, with the standard orientation cutting hair less short than said tight orientation, the method including the steps of: removing a head of the trimmer from a handle of the trimmer; disassembling the head of the trimmer into a separate fixed blade and a separate vibrating blade by sliding a block removably attachable to the vibrating blade laterally out of a slot in the fixed blade, extending parallel with a tip line of the fixed blade to which teeth of the fixed blade extend; detaching the

block from the vibrating blade; rotating the block 180°; reattaching the block to the vibrating blade; reassembling the head of the trimmer by sliding the block back into the slot of the fixed blade, with the tip line of the fixed blade and the tip line of the vibrating blade now closer to each other than they were previously; and reconnecting the head to the handle.
